Exp no: 06

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Error Correction at Data Link Layer

Aim:-

Write a program to implement error detection and correction using HAMMING code concept. Make a test run to input data stream and verify error correction feature.

Create Sender program with below features

1. Input to sender file should be a text of any length. Program should convert the text to binary.
2. Apply hamming code concept on the binary data and add redundant bits to it.
3. Save this output in a file called channel.

Create a receives program with below features

1. Receiver program should read the input from channel file.
2. Apply hamming code on the binary data to check for errors.
3. If there is an error, display the position of error
4. Else remove the redunbant bits and convert the binary data to asii and display the output.

Code:-

```
def hamming_encryption(data):
    m = len(data)
    r = 0
    while (2**r) <= (m+r+1):
        r += 1
    r -= 1
    n = m+r+1
    code = [0]*n
    j = 0
    for i in range(1,n):
        if (i & (i-1)) != 0:
            code[i] = data[j]
            j += 1
    for i in range(r):
        pos = 2**i
        pair = 0
        for j in range(1,n):
            if j & pos:
                pair ^= int(code[j])
        code[pos] = str(pair)
    return (code[1:])
def hammingcode_description(data, r, m):
    for i in range(r):
        pos = 2**i
        pair = 0
        for j in range(1, r+m+1):
            if j & Pos:
                pair ^= int(code[j])
```

receiver.py

```
if (pair != 0):
  if (data[pos] == '0'):
    data[pos] = "1"
else
  data[pos] = '0'
return
    join(data)
data = input("Enter binary data: ")
print("Data after encoding from sender side", hamming_encryption)
rdata = input("Enter data received by the receiver")
print(hammingcode_decryption(rdata, r, rdata length))
```

output:-

① Enter data bits: 1011100

Number of redundant bits (r) needed : 4

Total frame length (m+r) : 10

Data to be transmitted : 1011100011

Transmitted bit change? (yes/NO) : yes

Enter the corrupted data stream (10 bits) : 1011101011

Verification Result: Error detected!

Wrong bit is at position: 4

Corrected data streams: 1011100011

(2)

Enter data bits: 1011

number of redundant bits (r) needed: 3

Total data frame length (m+r): 7

Data to be transmitted: 1010101

Transmitted bit change? (yes/no): No

Verification Result: no error detected in transmission.

Result:-

Thus the hamming code was implemented successfully for error detection and correction.